

AI-Powered Industrial Chatbot Using Hugging Face Large Language Models

AI & ML Mini Project — Industrial Training Program on Robotics and AI

Student: _____

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1. Problem Statement

The project implements an AI-powered industrial chatbot that uses a Hugging Face language model, answers questions from technical documents, applies prompt engineering, and provides a usable web or command-line interface. The implementation follows the six tasks specified in the assignment brief.

2. System Architecture

The system follows a retrieval-augmented generation (RAG) workflow. Documents are converted into text chunks, embedded into a vector space, and searched using cosine similarity. The retrieved evidence is inserted into a controlled prompt and passed to the Hugging Face instruction-tuned LLM. The generated answer is displayed through Streamlit or the CLI.

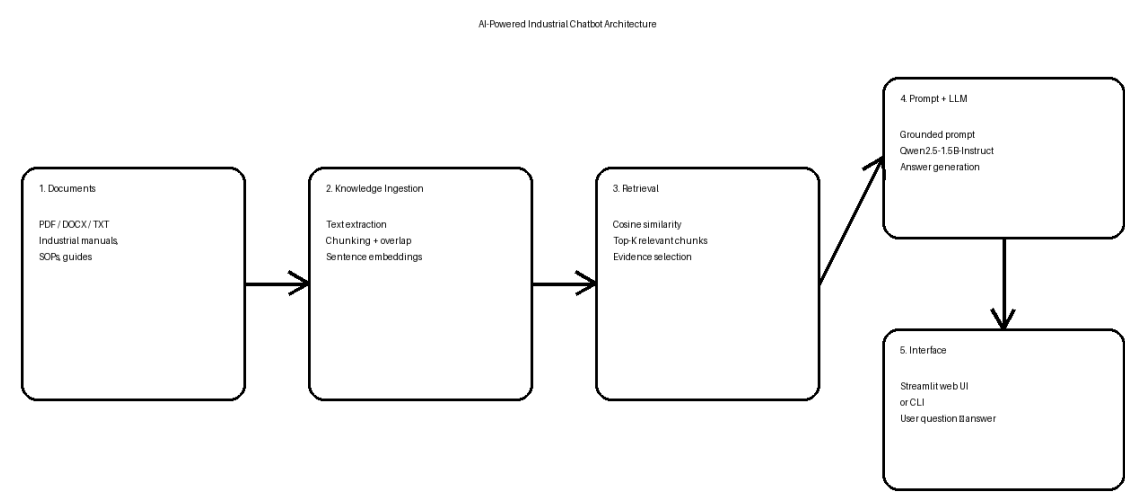


Figure 1. System architecture showing document ingestion, embeddings/retrieval, LLM generation, and user interface.

3. Model Selection

Component	Selected model	Purpose
LLM	Qwen/Qwen2.5-1.5B-Instruct	Instruction-following answer generation
Embedding model	sentence-transformers/all-MiniLM-L6-v2	Semantic document and query embeddings

Qwen2.5-1.5B-Instruct was selected because it is an instruction-tuned Hugging Face causal language model and its model card documents direct use with Transformers. The repository currently lists the model weights at about 3.09 GB, making the model practical for a local demonstration on a consumer GPU with sufficient memory. The embedding model maps sentences and paragraphs to a 384-dimensional dense vector space and is designed for semantic-search style applications.

The implementation intentionally avoids fine-tuning for this mini-project. RAG plus prompt engineering is faster to implement, keeps the source documents externally editable, and directly addresses the assignment's

requirement to improve domain-specific answers.

4. Knowledge Ingestion

The ingestion pipeline accepts TXT, Markdown, PDF, and DOCX files. Text is normalized and divided into overlapping chunks. Each chunk is embedded with all-MiniLM-L6-v2 and stored with its source filename and chunk identifier. Embeddings are normalized, so the dot product can be used as cosine similarity during retrieval.

For the demonstration package, three sample industrial documents are included: robot workcell safety, CNC machine operation, and PLC troubleshooting/maintenance. If the instructor supplied separate technical documents, they can be placed in the data/ directory and the index rebuilt with the same command.

5. Prompt Engineering

The system prompt defines the chatbot as an industrial technical support assistant and constrains it to the retrieved document context. It explicitly prohibits inventing specifications, limits, procedures, and safety requirements. When the retrieved evidence is insufficient, the model is instructed to return a fixed fallback statement. The prompt also asks for source references such as [Source 1], which improves traceability during a technical Q&A; interaction.

6. Q&A; Pipeline Implementation

1. Receive a user question.
2. Encode the question using the same embedding model used for the documents.
3. Calculate cosine similarity against all indexed chunks.
4. Select the top four chunks as evidence.
5. Insert the evidence into a grounded system/user prompt.
6. Generate the response with Qwen2.5-1.5B-Instruct.
7. Display the answer and retrieved evidence in the interface.

7. Chatbot Deployment

Two interfaces are provided. The CLI is launched with `python src/cli.py`. The web interface is launched with `streamlit run app.py`. The Streamlit application displays the conversation and exposes the retrieved document evidence for each answer.

8. Demonstration Questions

Question	Expected evidence source
What precautions are required before entering a robot workcell?	Industrial robot safety guide
What should be checked before restarting the CNC machine?	CNC operation and maintenance guide
What are symptoms of PLC communication failure?	PLC troubleshooting guide
What if the answer is not in the documents?	Grounding/fallback prompt behavior

9. Limitations

The demonstration is a small local RAG system rather than a production industrial control system. Retrieval quality depends on document quality, chunking, and embedding similarity. The LLM can still produce an incorrect answer if retrieved evidence is ambiguous, so generated answers should not be treated as a substitute for approved industrial procedures, manufacturer manuals, or safety controls.

10. Conclusion

The completed system satisfies the assignment requirements by combining Python, Hugging Face Transformers, a Hugging Face embedding model, document ingestion, retrieval, prompt engineering, question answering, and a usable CLI/Streamlit interface. The modular design also allows the knowledge base to be

replaced with instructor-provided industrial documents without changing the core chatbot architecture.

11. References

1. Hugging Face model card: Qwen/Qwen2.5-1.5B-Instruct. <https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct>
2. Hugging Face model card: sentence-transformers/all-MiniLM-L6-v2.
<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>
3. Assignment brief supplied for the project: AI-Powered Industrial Chatbot (Generative AI), pages 1–2.